

Dae Yeon Kim Ph.D.

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EDUCATION

Ph.D.	Seoul National University , Seoul, Korea School of Chemical and Biological Engineering Thesis: “A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow” (Advisor: Prof. Kyung Hyun Ahn)	Aug 2023
B.S.	Sogang University , Seoul, Korea Department of Chemical and Biomolecular Engineering Graduation honors, <i>Summa Cum Laude</i>	Feb 2017

PROFESSIONAL EXPERIENCES

Assistant Professor	Department of Chemical and Biomolecular Engineering Chonnam National University, Yeosu-si, Jeollanam-do, Korea	Mar 2026 – Present
Postdoctoral Scholar	Department of Chemical Engineering Stanford University, Stanford, CA, USA (Advisor: Prof. Sho Takatori)	Feb 2025 – Jan 2026
Postdoctoral Scholar	Department of Chemical Engineering University of California, Santa Barbara, Santa Barbara, CA, USA (Advisor: Prof. Sho Takatori)	Mar 2024 – Jan 2025
Postdoctoral Researcher	Center for Coating Materials and Processing Seoul National University, Seoul, Korea (Advisor: Prof. Kyung Hyun Ahn)	Sep 2023 – Jan 2024

RESEARCH EXPERIENCES

Research Skills

Microfluidics	• Design and fabrication of microchannel • Fluid visualization and PIV • Pressure drop measurement
Rheometry	• Rotational rheometer (ARES, AR-G2, DHR) • Extensional rheometer (Capillary Breakup Extensional Rheometer) • Passive/Active microrheology (Optical tweezer)
Capillary breakup	• Measurement of elongational property of complex fluids (polymer solution, colloidal suspension) • Direct visualization of thinning filament surface via high-speed camera
Optical tweezer	• Measurement of inter-particle interaction (e.g. Hydrodynamic interaction) • Microrheology of soft matter
Chemical labeling	• Soft vesicle labeling (SUV, GUV) • Colloids and bacteria labeling
Bacterial culturing	• Characterization and labeling • Quantitative imaging analysis

- Analysis tool
- Image processing and quantitative analysis (MATLAB, Python)
 - CFD simulation (COMSOL)

Research Interests

Complex materials and soft matter physics

- Colloidal suspension rheology
- Fluid-mediated interactions between colloids
- Active matter rheology and control
- Colloidal fouling onto a membrane surface (complex interactions between foulants and membrane)
- Flow instability of viscoelastic fluid

Heterogeneity control of industrial slurry (e.g. lithium-ion batteries, fuel cells, CMP slurry)

- Stability control of colloidal slurry
- Characterization of slurry for quality control

Data-driven machine learning classification

- Characterization of complex fluid
- Clogging and fouling prediction

Image-based rheological properties measurement

- Discovery of constitutive relations for complex fluids
- Dynamics of soft materials with active forcing

PUBLICATIONS

Peer-reviewed Journal Articles

* equal contribution † co-corresponding

13. Haegyo Jeong, Minsun Hwang, **Dae Yeon Kim**[†], Seon Yeop Jung[†], Influence of fluid viscosity and cavity length on particle transport and deposition in expansion-contraction microchannels, *Korea-Australia Rheology Journal*, **2026**, 38, 335–354. [\[Link\]](#)
12. **Dae Yeon Kim**^{*}, Sachit G. Nagella^{*}, Saksham Malik, Nayeon Park, Jaewook Nam, Eric S.G. Shaqfeh, Sho C. Takatori, Colloidal hydrodynamic interactions in viscoelastic fluids, *Soft Matter*, **2025**, 21, 8035–8048. [\[Link\]](#)
11. Minsun Hwang, **Dae Yeon Kim**, Kyung Hyun Ahn, Sung Sik Lee, Seon Yeop Jung, Particle deposition in colloid-polymer suspensions: Complex interplay between flow characteristics and polymer interactions, *Physics of Fluids*, **2025**, 37, 073389. [\[Link\]](#)
10. **Dae Yeon Kim**^{*}, Sachit G. Nagella^{*}, Kyu Hwan Choi, Sho C. Takatori, Direct experimental measurement of many-body hydrodynamic interactions with optical tweezers, *Physical Review Fluids*, **2025**, 10, 064301. [\[Link\]](#)
9. Woobin Yi^{*}, **Dae Yeon Kim**^{*}, Howon Jin, Sangwoong Yoon, Kyung Hyun Ahn, Early detection of pore clogging in microfluidic systems with 3D convolutional neural network, *Separation and Purification Technology*, **2025**, 359, 130428. [\[Link\]](#)
8. Young Jin Lee, **Dae Yeon Kim**, Kyung Hyun Ahn, Particulate fouling simulation in unit micropore using a hydrodynamically coupled Lagrangian framework, *Desalination*, **2025**, 593, 118255. [\[Link\]](#)
7. Minsun Hwang, **Dae Yeon Kim**, Seon Yeop Jung, Kyung Hyun Ahn, Effect of fluid viscosity on the deposition of colloidal particles in an expansion-contraction microchannel flow, *Nihon Reorogi Gakkaishi*, **2024**, 52 (4), 231–240. [\[Link\]](#)

6. Wonsik Shin, Joung Sook Hong, **Dae Yeon Kim**, Si Yoon Kim, Kyu Hyun, Jun Dong Park, Kyung Hyun Ahn, Effect of cellulose nanocrystals on the emulsion stability and rheological properties of microalgal Pickering emulsions, *Algal Research*, **2024**, 83, 103731. [\[Link\]](#)
5. Young Jin Lee, Howon Jin, **Dae Yeon Kim**, Seunghoon Kang, Kyung Hyun Ahn, Lagrangian-based simulation method using constrained Stokesian dynamics for particulate flows in microchannel, *Journal of Fluid Mechanics*, **2024**, 990, A3. [\[Link\]](#)
4. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Howon Jin, Kyung Hyun Ahn, From successive deposition to clogging of poly(styrene) particles in the cross-flow through a T-shaped microchannel, *Separation and Purification Technology*, **2023**, 322, 124263. [\[Link\]](#)
3. **Dae Yeon Kim**, Seon Yeop Jung, Young Jin Lee, Kyung Hyun Ahn, Effect of colloidal interactions and hydrodynamic stress on particle deposition in a single micropore, *Langmuir*, **2022**, 38 (19), 6013–6022. [\[Link\]](#)
2. **Dae Yeon Kim**, Kyung Hyun Ahn, Effect of operating conditions on the sessile drop oscillation of PEO solution in the capillary thinning process, *Nihon Reorogi Gakkaishi*, **2021**, 49 (2), 127–133. [\[Link\]](#)
1. Youngseok Kim, **Dae Yeon Kim**, Joung Sook Hong, Kyung Hyun Ahn, Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow, *Langmuir*, **2018**, 34 (38), 11454–11463. [\[Link\]](#)

Submitted/in-progress

3. **Dae Yeon Kim**, Free-vibration mechanism and viscoelastic damping of sessile-drop oscillation in capillary thinning, *in preparation*.
2. **Dae Yeon Kim**, Colloidal hydrodynamic interactions: from bulk rheometry to rotation-based optical tweezer measurements, *under review*.
1. **Dae Yeon Kim**, Izzie Allydice, Sho C. Takatori, Colloidal hydrodynamic interaction in bacterial active bath, *in preparation*.

PRESENTATIONS

International Conference

9. Fluid-mediated rotational coupling and hydrodynamic interactions in colloidal particles using optical trapping, *APS Global Physics Summit (APS March Meeting 2025)*. 16 – 21 Mar 2025, Anaheim, CA, USA.
8. Successive deposition of poly(styrene) particles in cross-flow through T-shaped microchannel, *The 19th International Congress on Rheology (ICR 2023)*. 29 July – 4 Aug 2023, Athens, Greece.
7. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 8th Pacific Rim Conference on Rheology (PRCR 2023)*. 15 – 19 May 2023, Vancouver, BC, Canada.
6. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *The 93rd Annual Meeting of the Society of Rheology (SoR 93rd Annual Meeting)*. 9 – 13 Oct 2022, Chicago, IL, USA.
5. Effect of salt concentration and flow rate on clogging dynamics at the single micro-pore. *11th Korea-Australia Rheology Conference (KARC 2021)*. 24 – 25 Nov 2021, Virtual.

4. The effect of operating condition and rheological properties of PEO solution on the sessile drop oscillation during elasto-capillary thinning. *International Union of Materials Research Societies-International Conference in Asia (IUMRS-ICA 2020)*. 23 – 26 Feb 2021, Virtual.
3. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *The 15th International Workshop for East Asian Young Rheologists (IWEAYR 2020)*. 8 – 10 Jan 2020, Changchun, China.
2. Effect of rheological properties of PEO solution on the sessile drop oscillation during the early stages of elasto-capillary thinning transition. *Annual European Rheology Conference (AERC 2019)*. 8 – 11 Apr 2019, Portoroz, Slovenia.
1. Agglomerate breakup of destabilized polystyrene particles under a cross-channel planar extensional flow. *The 14th International Workshop for East Asian Young Rheologists (IWEAYR 2019)*. 23 – 26 Jan 2019, Nagoya, Japan.

Others (2023 – Present)

5. Particle Deposition, Pore Clogging, and AI-Assisted Prediction of Membrane Fouling. *KSIEC Special Symposium: AX in the Carbon-Neutral Era, Korean Society of Industrial and Engineering Chemistry*. 10 – 11 Sep 2026, Yeosu, Korea
4. Probing hydrodynamic interactions and rheology in complex fluids using optical laser tweezers. *2026 Annual Meeting of the Korean Society of Rheology*. 7 – 8 May 2026, Seoul, Korea
3. Prediction of membrane fouling via particle transport phenomena and ai-based analysis. *2026 Spring Conference of the Korean Society for Energy*. 15 – 17 Apr 2026, Yeosu, Korea
2. Direct experimental measurement of hydrodynamic interactions using optical tweezers. *Shaqfeh Group Seminar, Stanford University*. May 2025
1. A study on particle deposition and clogging dynamics of colloidal suspensions in complex flow. *EEPL Group Seminar, Dankook University*. Aug 2023

TEACHING AND MENTORING

Teaching (S:Spring, F:Fall)	Transport Phenomena (Chonnam National University)	2026F
	Chemical Reaction Engineering II (Chonnam National University)	2026F
	General Chemistry I (Chonnam National University)	2026S
	Chemical Reaction Engineering (Chonnam National University)	2026S
	Creative Academy: Basic Chemical Engineering (Janghoon High school, Seoul)	Mar 2018 – Jan 2019
TA	Transport Phenomena (Seoul National University)	Spring. 2018
	Introduction to Chemical Engineering (Seoul National University)	Fall. 2019
Mentoring	2025 ChemE REU Program	Jul 2025 – Present
	Mentee: Izzie Allydice (Stanford '28)	
	EEPL Group, Dankook University	Mar 2022 – Jul 2025
	Mentee: Minsun Hwang (B.S.–M.S. Integrated Program)	

AWARDS AND HONORS

Best Paper Award	DONGJIN Best Paper Award (supported by DONGJIN SEMICHEM CO., LTD.)	Jul 2022
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Best Presentation Award	Best Presentation Award of 15th IWEAYR (The 15th International Workshop for East Asian Young Rheologists)	Jan 2020
Academic Scholarship	Scholarship for academic excellence (Sogang University)	Sep 2013 – Aug 2016

OTHER ACTIVITIES

Member	Young Engineers Honor Society (YEHS)	Jan 2014 – Present
Student Leader	Department of Chemical and Biomolecular Engineering Sogang University, Korea	Jan 2014 – Mar 2014

REFERENCES

Dr. Kyung Hyun Ahn	Professor, School of Chemical and Biological Engineering, Seoul National University, Seoul, Korea	ahnnet@snu.ac.kr
Dr. Sho Takatori	Associate Professor, Department of Chemical Engineering, Stanford University, Stanford, CA	stakatori@stanford.edu
Dr. Seon Yeop Jung	Assistant Professor, Department of Chemical Engineering, Dankook University, Gyeonggi-do, Korea	seon27@dankook.ac.kr